VaaniShield

Technical Requirements Document (TRD)

AI Project Submission — TRD
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1. System Architecture Overview

The pipeline runs in five stages: (1) transcript-based code-switch localization, (2) boundary-local feature extraction around each switch point, (3) a parallel speaker-nulled global artifact stream from a self-supervised speech embedding, (4) late fusion of both streams into a single classifier, and (5) an explainability layer that attributes the score back to specific code-switch points. Stages 1–3 run independently per utterance and are combined only at the fusion step, so each stream's contribution can be evaluated in isolation.

2. Data Requirements & Dataset Construction

Component	Specification
Bona-fide code-switched speech	Hindi-English (primary) + one additional regional-language pair; consenting speakers + public corpora (IndicTTS, Common Voice Indian subsets).
Cloned/spoofed speech	Same speakers/scripts re-synthesized via ≥3 open voice-cloning systems (Coqui XTTS, Bark, OpenVoice) to avoid single-synthesizer bias.
Code-switch annotation	Token-level language ID + forced alignment (Montreal Forced Aligner or Whisper-derived alignment) to timestamp each switch point.
Noise/channel augmentation	Codec compression, packet-loss simulation, background noise injection, applied to both bona-fide and spoofed classes equally.
Held-out generalization set	At least one voice-cloning system excluded from training, used only for the generalization evaluation.

3. Module Specifications

3.1 Code-Switch Boundary Localization

Input: audio + transcript. Process: per-token language tagging → forced alignment for timestamps → flag consecutive tokens with differing language tags as switch points. Output: list of switch-point timestamps per utterance, each with a ±200–400 ms window.

3.2 Boundary-Local Feature Extraction

Input: audio window around each switch point. Features: F0/pitch reset magnitude, energy discontinuity, phone-duration ratio vs. speaker baseline, formant-transition smoothness, frame-to-frame MFCC delta magnitude (via librosa/Praat). Output: a fixed-length feature vector per switch point, aggregated (e.g. mean/max) to one vector per utterance.

3.3 Speaker-Nulled Global Stream

Input: full utterance audio. Process: extract a WavLM/wav2vec2 embedding → apply an orthogonal speaker-subspace projection (SNAP-style) to suppress speaker-identity components. Output: a speaker-invariant artifact-residual embedding.

3.4 Fusion & Classification

Input: concatenated boundary-local vector + speaker-nulled embedding. Model: logistic regression or a small MLP (kept simple deliberately, for interpretability in ablation). Output: fraud-risk probability + bootstrap confidence interval + per-switch-point attribution for the explainability layer.

4. Technical Stack & Dependencies

Layer	Tools / Libraries
Audio processing	librosa, Praat/parselmouth, torchaudio
Speech embeddings	WavLM / wav2vec2 (HuggingFace transformers)
Forced alignment	Montreal Forced Aligner or Whisper-derived alignment
Voice cloning / TTS (for spoof generation)	Coqui XTTS, Bark, OpenVoice
Classifier	scikit-learn (logistic regression) or a small PyTorch MLP
Explainability	SHAP (feature attribution over the fused vector)
Evaluation	EER computation utilities, bootstrap CI (numpy/scipy)
Demo	Streamlit

5. Evaluation Methodology & Metrics

- **Baselines:** Baseline A (generic WavLM/wav2vec2 classifier, no boundary-awareness/speaker-nulling), Baseline B (re-implementation of the published Mel+LFCC+Wav2Vec Indian-accent fusion model).
- **Ablations:** boundary-local stream alone vs. speaker-nulled stream alone vs. full fusion.
- **Metrics:** equal error rate (EER), precision/recall, bootstrap confidence intervals; reported per-language-pair and per-noise-condition rather than pooled.
- **Generalization test:** evaluate on the held-out voice-cloning system excluded from training.

6. Non-Functional Technical Requirements

- Feature-extraction and embedding steps must run on commodity GPU/CPU available within the academic compute budget (no large-scale distributed training required).
- All dataset-generation and training code must be scripted (not manual/one-off) so results are reproducible.
- Demo response time: a few seconds per short call clip is acceptable; strict real-time/on-device latency is explicitly not a requirement for this submission (see PRD, Section 5, Out of Scope).

7. Security & Privacy Considerations

- Only consenting speakers' recordings are used for bona-fide and cloning-source audio; no real intercepted fraud-call audio is collected or used.
- Generated cloned audio is used strictly for research evaluation and is not distributed outside the project's academic use.
- Any demo deployment should avoid persisting uploaded call audio beyond the session unless explicit consent is given.

8. Technical Risks & Mitigations

Risk	Mitigation
Forced alignment/language-ID errors mislabel code-switch points.	Manually spot-check a sample of annotated switch points before full-scale feature extraction.
Boundary-local features are noisy in low-quality/compressed audio.	Evaluate under multiple noise conditions explicitly, rather than assuming clean-audio performance transfers.
Speaker-nulling projection removes too much signal, hurting accuracy.	Ablate speaker-nulled stream alone vs. raw embedding to quantify the trade-off directly.